

EMA Leads Funnel — Change Report

Referral tunnel build, internal rebrand & production rollout

Reporting window: 3 July 2026 → 4 July 2026 (last 24 hours) · Prepared 4 July 2026

1. Executive summary

Over the last 24 hours the **EMA Leads Funnel** (our lead-capture funnel that feeds the separate main E-Migration Assist platform) received two connected pieces of work, both now **complete, tested, and published live**:

DONE A. Internal-only rebrand. Our internal/operational name is now "EMA Leads Funnel", while the public brand our leads and visitors see stays "E-Migration Assist" everywhere.

DONE B. Referral tunnel (sender side). With a lead's explicit consent, the funnel can now offer a *privacy-redacted* referral to a matched partner firm and hand it off securely to the main EMA platform.

The funnel side is fully verified on both our development and live environments. The **only remaining item is on the other project (the main EMA platform)**: its copy of the shared security key must be made identical to ours — see §6.

12

COMMITTS

25

FILES CHANGED

~1,800

LINES ADDED

1

NEW DB TABLE

2. Background — what this funnel is

This repository is the **lead-capture funnel**, not the main EMA platform. It captures prospective clients through a guided assessment and manages them in an admin CRM. The new referral capability lets qualified, consenting leads be passed to a partner law/immigration firm, who then picks them up inside the *separate* main EMA system. We built only the **sending** side of that hand-off; the receiving side lives in the other project.

3. Part A — Internal-only rebrand

We separated our *operational* identity from our *public* identity, so the team and admin tools reflect the funnel's true role without confusing the people we serve.

Now says "EMA Leads Funnel" (internal)	Still says "E-Migration Assist" (public)
Repository README, project description, admin browser tab titles, admin logo alt-text, internal team notifications	Assessment / capture pages, referral preview shown to firms, all customer emails (incl. the firm offer sign-off), WhatsApp messages

Guiding principle recorded for the team:*go by role, not name*— this repo is the funnel, the public brand stays intact.

4. Part B — Referral tunnel (what it does)

The tunnel adds a consent-based referral pathway with privacy and safety built in from the ground up.

How it flows

- **Consent:** a lead explicitly opts in (POPIA-aligned) to be shared with a partner firm.
- **Match & offer:** the funnel matches a suitable firm and offers them a *redacted* summary (matter type, urgency, region — **no name, email, or phone**).
- **Accept & open:** the firm accepts and is redirected into the main EMA platform via a short-lived, signed link; the applicant's full details are delivered securely to EMA at that moment.
- **Status callbacks:** EMA reports progress back (account linked, converted, failed), which the funnel records.
- **Admin board:** a new `/admin/referrals` view lets the team see referral status.

Safety guarantees engineered in

- **Personal data is structurally protected.** The referral records table has *no* name/email/phone columns at all — personal details only ever travel inside a single encrypted, signed hand-off to EMA.
- **Every message is cryptographically signed** (HMAC-SHA256) with a shared secret, so neither side can be spoofed.
- **Fails safely.** If the shared secret is missing, the tunnel refuses to operate rather than leak anything (returns a "not configured" response).

- **No duplicates, no data loss.** Consent creation locks the lead so two clicks can't create two referrals; a "converted" referral can never be accidentally regressed to an earlier state.

5. Publishing & verification

The funnel was **published live** during this window and independently verified end-to-end.

Test performed (run against the live/dev funnel)	Result
Lead consents → referral created & firm matched	Pass
Repeat consent → same referral, no duplicate	Pass
Firm preview shows redacted info only (no personal data)	Pass
Accept → secure signed redirect into EMA	Pass
Callback with wrong signature → rejected	Pass
Callback with valid signature → accepted	Pass
"Converted" status held against a late/regressing message	Pass
Missing secret → tunnel refuses to send (fail-safe)	Pass
Referral records confirmed to contain no personal-data columns	Pass
Live published funnel reachable & verifying signatures correctly	Pass

6. Open item — action on the main EMA platform (not the funnel)

ACTION NEEDED

During cross-system testing, the main EMA platform's signed status callback was rejected by the funnel with a signature error. We proved conclusively this is **not a funnel problem**: a callback signed with the funnel's own key passes the live funnel's signature check, so the funnel's development and production keys match each other and the contract is correct.

Root cause: EMA's copy of the shared secret (`REFERRAL_TUNNEL_SECRET`) is not byte-identical to the funnel's — a copy-paste drift (typically a stray space or newline). **Fix (on the EMA side):** re-paste the exact same secret value into EMA's workspace and production secrets, then republish EMA. No funnel change required.

7. Configuration status & go-live checklist

Item	Side	Status
Shared secret <code>REFERRAL_TUNNEL_SECRET</code>	Funnel	Set (dev + production match)
EMA target URL <code>EMA_APP_URL</code>	Funnel	Set to EMA dev URL (swap to EMA's live URL after EMA is published)
Match shared secret exactly	EMA	Pending re-paste (see §6)
Point <code>FUNNEL_API_URL</code> at the live funnel	EMA	Use <code>https://immigrationassist.replit.app</code>
Enable referrals flag <code>FF_PARTNER_REFERRALS=true</code>	EMA	Pending (EMA production)
Publish the main EMA platform	EMA	Pending

Once EMA's secret matches and EMA is published, the round-trip is live. The single remaining funnel-side step is repointing `EMA_APP_URL` from EMA's dev address to its published address.

Appendix A — Commit history (last 24h)

Commit	When	Summary
94df561	07-03 20:02	Add instructions for building the referral tunnel sending side
46f8efa	07-03 20:10	Add clarification document explaining referral tunnel system setup
c9c3867	07-03 20:56	Add a referral tunnel for lead management and firm matching
d0de6db	07-03 21:10	Improve referral system security and update application branding
38acd4b	07-03 22:54	Add new image assets for user interface elements
e0df3ad	07-03 23:00	Document the referral tunnel contract and rebrand details
5c62257	07-03 23:23	Published the App (live)
0bbde23 - 995ce46	07-03 23:32 → 07-04 00:03	Coordination screenshots & asset updates (secret-parity investigation)

Appendix B — Files changed (25 total)

Backend — new tunnel engine (api-server)

- lib/referralTunnel.ts — signing/verification primitives (the security contract)
- lib/referralService.ts — firm matching, redacted preview, applicant hand-off, token minting
- lib/referralAudit.ts — append-only audit trail
- routes/referrals.ts — consent, preview, admin referral endpoints
- routes/referralGate.ts — accept/redirect + status callback receiver
- routes/index.ts , routes/support.ts — wiring

Database

- lib/db/src/schema/referrals.ts — new referrals table (privacy-safe: no personal-data columns) + audit table
- lib/db/src/schema/index.ts , drizzle.config.ts — registration

Frontend (emigration-assist)

- src/components/referral-consent-card.tsx — lead consent UI
- src/pages/referral-preview.tsx — redacted preview shown to firms
- src/pages/admin-referrals.tsx — admin referral board
- Rebrand touches: App.tsx , admin-layout.tsx , admin-user-menu.tsx , admin-login.tsx , admin.tsx , admin-lead-detail.tsx , thank-you.tsx , index.css , package.json , artifact.toml , opengraph.jpg

Docs

- replit.md — architecture decision record for the tunnel & rebrand
- .agents/memory/ — durable contract notes for future work

Appendix C — New API endpoints

Endpoint	Purpose
POST /api/referrals/consent	Lead opts in; creates & matches referral
GET /api/referrals/preview/:id	Redacted preview for the firm
GET /api/referral-gate/redirect/:id	Accept → secure signed redirect into EMA
POST /api/referral-gate/callback	Receives signed status updates from EMA
GET /api/admin/referrals*	Admin listing/detail

EMA Leads Funnel — internal change report · Generated 4 July 2026 · Live backend: [immigrationassist.replit.app](#) · This document summarises work on the funnel (sender) only; the main E-Migration Assist platform is a separate project.